

CORNELL MASTER BEEKEEPING - INSPECTIONS

OBJECTIVE / TOOL KIT / WEATHER

POPULATION STATUS

1. **Population:** How many adult bees are in the colony? Is the number appropriate for the season?
2. **Queen status:** Is the queen laying worker eggs? Is there evidence that the bees are intending to supersede her or swarm? Is there evidence of laying workers?
3. **Brood quality and quantity:** Is the brood solid or spotty? Is it sufficiently abundant for the time of year? Is brood of all ages present — eggs, larvae, and capped brood?
4. **Population dynamics:** If you compare the apparent brood rearing activity with the natural attrition rates of foragers at this time of year, does the population seem to be increasing or decreasing?
5. **Drone population:** Is drone production seasonally appropriate? Are there a disproportionate number of drones?

NUTRITION

1. **Floral sources:** What flowers are blooming in your area at the moment, and what pollen sources are your bees bringing in? Are there environmental conditions that may be impacting the bees' nutrition, e.g., prolonged bad weather or a drought?
2. **Foraging behavior:** Are there sufficient numbers of bees foraging? Seeing 10 to 20 bees entering with full pollen baskets per minute is an indication that lots of resources are coming into the colony. Are bees bringing in pollen or propolis in their pollen baskets or nectar/water in their honey stomachs?
3. **Food stores in the colony:** How much pollen/bee bread and nectar/honey is stored? Is this amount appropriate for the season?
4. **Larval nutrition:** Are the young larvae floating in abundant pools of royal jelly?

PESTS PARASITES & DISEASES

1. Do you see any **symptoms of brood disease** in the colony? Not all problems will be visible, but brood that is discolored, twisted, melted-looking, partially cannibalized, or has punctured cappings may indicate a problem.
2. Do you see any **adult bees with deformities or behavioral abnormalities**? Do you see any *Varroa* mites or pests such as hive beetles, wax moths, or the larvae of beetles and moths? It is important to note that for *Varroa*, without sampling procedures that follow specific protocols, such observations alone do not necessarily provide a full picture of the mite pressure within a colony.
3. Do you see **other pests** — or evidence of such pests — present in the honey supers or other areas of the hive?
4. Are the **bees acting more defensively** than usual?

Outside the Hive

PREDATOR EVIDENCE

Check the outside of the hive for damage or signs of predators. Skunks often scratch the ground in front of a hive or at the entrance to lure out curious bees. Bees that come to check out the sound get eaten. In this way, skunks can hasten the depopulation of weak colonies. If you see scratch marks on the ground or your hives, elevate your hive on a stand so the skunks can no longer reach the entrance. You can also put small mammal fencing around your apiary to keep them out.

FECAL MATTER

Look at the exterior of the hive for bee fecal matter. Bees defecate outside the colony to maintain a clean home and reduce the spread of infection to their nestmates. A little bit of fecal matter on the outside of the hive is normal; in fact, some fecal matter visible on snow in winter is a good sign, since it means that the bees found a mild day to defecate outside after a long period of confinement. But excessive drips of orange or brown stains on the front of a hive are a signal that something is wrong.

GUARD BEES

The last thing to look for before you open the colony is guard bees. They stand at the entrance, often holding their front legs up. They check out bees entering the hive — and may even be checking you out. Wave your hand slowly in front of the entrance. If a guard bee follows your hand, she is watching you. You'll need to use smoke to break her attention.

Inspecting Honey Supers

Adding Supers

Looking for Signs of Pests

Assessing Moisture

Understanding Nectar Flows

Starting Your Inspection of the Brood Nest

Inspect at least 4 frames of brood per box

Adult bees

Is the adult **population strong**? If there are sufficient numbers of bees to nearly cover both sides of each frame in the brood nest, the colony has a strong population. In contrast, if there are not enough bees to cover the brood, it may signal a problem and cause other issues. For instance, the brood is at risk of being chilled in a colony with a low population.

Next, look for **indications of diseases** on the adult bees, such as hairlessness, greasiness, deformed wings, or twitching movements. Estimate the proportion of drones in the adult population: too many may indicate you have a drone-laying queen or laying workers. During peak drone production in late spring and early summer, about 5% of the adult population in a hive will be drones.

As you inspect the frames, scan for the queen to see if she has a retinue of bees responding to her pheromones. Work to make sure she doesn't get harmed during the inspection. If you plan on pulling more frames or doing anything that might stress or harm her (e.g., sampling bees for mites), you can ensure her safety by gently caging her and placing her in a shady spot while you work, and reintroducing her to the top brood chamber at the end of the inspection. If you cage her, be careful not to forget about her!

Brood Health

If you have a strong population with many adults covering the frames, it may be difficult to assess the brood pattern beneath the bees. In these cases, you can gently blow on them or place your hand over them so that they move out of the way, or shake the bees off into the hive body you are working in. When inspecting brood frames, hold the frame with the sun at your back and angle it so the sun shines directly into the cells. A healthy brood frame has a **solid pattern of brood** (with few open cells), with **arcs of pollen and then honey** surrounding the brood. The frame will resemble a rainbow, with brood in the center, pollen arcing around it, and honey as the topmost layer. Some empty cells are normal: **6% to 17% of larvae are generally removed** in a healthy colony

(those that inherit identical same sex determination gene versions), but a higher percentage than that can indicate that the queen is poorly mated, inbred, or failing from age, or that the brood may be diseased or afflicted by parasites.

Diseased brood can take on a variety of symptoms depending on the parasite or pathogen in question. Instead of pearly white larvae, **diseased larvae may be coffee-colored, brown, gray, black, or appear chalk-like**. Their position may also change from being curled up nicely in their **cells to being twisted, pointy, melted, or mummified**. Infected pupae show other symptoms: **perforated or concave pupal cappings** are indicative of diseases, as are pupae that have been **uncapped and chewed down by nurse bees**. Lastly, bees that die upon emergence with their **proboscises extended** are indicative of some diseases and pesticide poisonings. Always be on the lookout for any signs of brood disease, even if symptoms are seen in only one cell. Take photographs and notes of any issues.

If the disease is unfamiliar to you or resembles American Foulbrood, send samples for free analysis to the Beltsville Bee Lab in Maryland, and label the affected colony so you can identify it later. Lastly, check for deposits in empty cells in the brood frame. Take note of any black scale that does not scrape off with a toothpick (indicative of American Foulbrood) or small patches of white crystals on the ceilings of brood cells called guanine; these crystals are indicative of high *Varroa* mite levels.

Brood Pattern and Sex

The queen's pattern of laying is important. A healthy hive has **brood of all ages: eggs, larvae, and capped pupae**. A good fertile queen will usually begin laying eggs near the center of the frame, and work her way outward in **concentric circles**. The result is brood **aging in a “step down” pattern**: larvae the same age sit next to each other, and moving outward from the center, the larvae are one day younger, then two days younger, and so on. As the fully developed bees emerge, the queen should fill those newly open cells with eggs. If you cannot find any eggs after a thorough inspection, you should take measures to replace the queen or introduce a frame with eggs on it so the bees can rear their own queen.

The **sex of the brood is important**. The colony starts investing in drone production once its population reaches around 4,000 individuals. For colonies of this size or bigger, **drones are most plentiful during swarm season**, when males are needed to mate with the newly emerging queens. Here in the Northeast, swarm season peaks from about mid-May to mid-July. **Drone production ceases toward the end of summer** as the colony begins to invest most of its energy into preparing for winter by building up its honey stores. On average, about **17% to 20% of comb in a colony is drone comb** (about two full frames' worth in a 10-frame hive body). This is the “natural” average in wild colonies and in foundationless hives, but beekeepers will likely find less drone comb in their hives if they are using frames of worker cell foundation. Drone comb will usually be built along the bottom or sides of the frames, not usually in the center of the brood nest. Drone comb can also be found in the burr comb on the tops of the frames or in other areas. If you see a high number of capped drone cells late in the summer (representing well over 5% of the drone population) or in colonies with weak populations, those colonies may have a drone-laying queen. A drone-laying queen will still lay only one egg per cell, while laying workers will lay multiple eggs in the cells.

Queen Cups and Queen cells

Examine the frames for queen cups and cells. **Queen cups are, by definition, small wax cups that are empty**. It is normal to find a few of these throughout the year, as the colony keeps them on hand in case it needs to raise a new queen or swarm. a couple queen cups do not signal any impending changes in the colony. This is true even if you find them in different places in the nest, as they may be torn down and rebuilt several times throughout a season. However, colonies will ramp up the production of queen cups in preparation of swarming. In this case, you can see many queen cups, usually close to a dozen, usually located along the lower edge of the brood nest. If you see **an egg or larva in the cup, it is now a queen cell**, indicating that the bees are seeking to raise a new queen. The placement of the cell can indicate why.

Generally, many queen cells along the **bottom of a frame signal that the colony is preparing to swarm**, and are known as swarm cells. In contrast, a few queen **cells near the center of the frame** are usually either queen **supersedure cells or emergency** queen cells. These can be used to replace an existing queen that may be failing, or to make up for the loss of a queen. The presence of queen supersedure cells is usually a sign that there is an issue with the queen and the workers are naturally replacing her.

Brood Nutrition

Take note of the brood nutrition. **One to two frames' worth of pollen/bee bread in each brood box** is ideal for brood rearing. However, some colonies may not always have that much on hand because they are using it up relatively quickly after they collect it. Observe the **diversity of pollen in the colony**. Does it appear that bees are foraging on pollen from a single source, or are there multiple sources of pollen (evident by various colors) present in the hive? Document any observations of **entombed pollen**, where bees cover a cell containing pollen with propolis and wax. Entombed pollen has been found to contain high levels of certain pesticides.

Besides looking at bee bread, you can get a good idea of a colony's nutritional levels based on **the amount of royal jelly surrounding young larvae**. Abundant white pools of royal jelly signal that there is a lot of protein available for nurses to synthesize into royal jelly, whereas colonies with little or no jelly visible around the larvae may benefit from receiving a protein supplement.

Condition of frames

Lastly, examine the condition of the comb. **Is the comb old and/or dark?** An old comb can harbor old pupal cocoons, disease spores, and pesticides, with the latter two posing a risk to colony health. Each year, aim to **replace your two oldest frames** in each hive body to keep your hives in optimum health. **However, do not remove more than four frames from a hive at once**, as the Bee Informed Partnership found this can have negative effects on the colony's population. Removing more than four frames from a hive at once will divert the bees' resources to building comb rather than raising brood or producing honey. There is a tradeoff between reducing the risks to the hive by discarding old, possibly contaminated, comb and requiring the bees to expend more of their energy.

Once the Colony Inspection Is Complete

As you finish inspecting frames, place them back in the boxes **in the same orientation and order** that they were in to ensure that the brood nest is intact — again, this is easily accomplished if you only take one to two frames out at a time. The inspection is complete when you have accomplished everything you set out to do. After the inspection, consider what sort of management the colony needs: do the bees need more room to store honey, or do you need to take steps to prevent or manage swarming? **Take note of when you need to return to your colonies for the next routine inspection or to follow up on anything you saw or did today.**

IDENTIFYING FRAMES

1	2	3	4	5	6	7	8	9	10
1	2	3	4	5	6	7	8	9	10